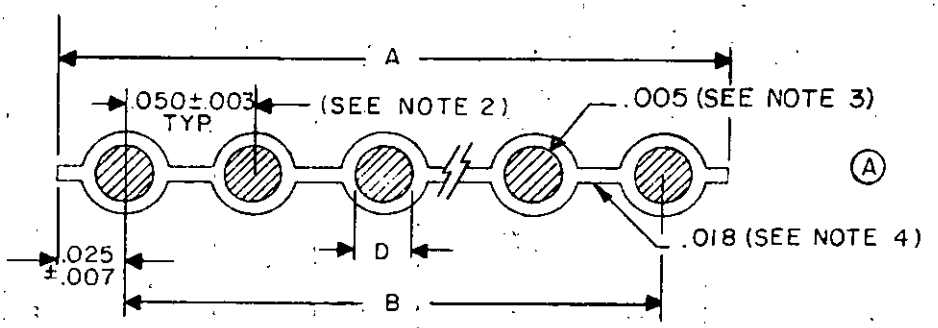
MILITARY SPECIFICATION SHEET

CABLES, SPECIAL PURPOSE, ELECTRICAL, FLEXIBLE, FLAT, UNSHIELDED, ROUND CONDUCTOR, 150°C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-49055.



INCHES	MM
.003	0.08
.005	0.13
.007	0.18
.018	0.46
.025	0.64
.050	1.27

Ⓐ NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Minimum insulation thickness.
4. Maximum web thickness.

FIGURE 1. Round conductor flat cable.

Ⓐ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

Ⓐ TABLE I. Dash numbers and characteristics.

Dash no.	No. of conductors	Cable width "A" inches	Cable thickness	Weight (lbs./1,000 ft. Nominal 1/	Weight (lbs./1,000 ft. Maximum	Conductor AWG	Conductor Dia "D" inches	Conductor Stranded	DC resistance at 20°C Ω /1000 ft	End conductor to end conductor "B" inches
01	9	.450 ± .010	.029 ± .003	10.67	12.10	28	.015	7/36	65,9	.400 ± 0.007
02	10	.500 ± .010	"	12.30	14.00	"	"	"	"	.450 ± 0.007
03	14	.700 ± .010	"	17.52	19.99	"	"	"	"	.650 ± 0.007
04	15	.750 ± .010	"	19.15	21.89	"	"	"	"	.700 ± 0.007
05	16	.800 ± .011	"	20.14	22.99	"	"	"	"	.750 ± 0.011
06	20	1.000 ± .011	"	25.36	28.97	"	"	"	"	.950 ± 0.011
07	24	1.200 ± .011	"	30.29	34.57	"	"	"	"	1.150 ± 0.011
08	25	1.250 ± .011	"	31.92	36.47	"	"	"	"	1.200 ± 0.011
09	26	1.300 ± .011	"	33.20	37.95	"	"	"	"	1.250 ± 0.011
10	34	1.700 ± .015	"	43.64	49.91	"	"	"	"	1.650 ± 0.015
11	37	1.850 ± .015	"	47.25	54.01	"	"	"	"	1.800 ± 0.015
12	40	2.000 ± .015	"	51.48	58.89	"	"	"	"	1.950 ± 0.015
13	50	2.500 ± .015	"	62.88	72.21	"	"	"	"	2.450 ± 0.015
14	60	3.000 ± .020	"	77.60	88.81	"	"	"	"	2.950 ± 0.015
15	64	3.200 ± .020	"	81.72	93.42	"	"	"	"	3.150 ± 0.015
16	9	.450 ± .010	.032 ± .003	13.77	15.20	26	.019	7/34	41,2	.400 ± 0.007
17	10	.500 ± .010	"	15.40	17.10	"	"	"	"	.450 ± 0.007
18	14	.700 ± .010	"	21.86	24.33	"	"	"	"	.650 ± 0.007
19	15	.750 ± .010	"	23.80	26.54	"	"	"	"	.700 ± 0.007
20	16	.800 ± .011	"	25.10	27.95	"	"	"	"	.750 ± 0.011
21	20	1.000 ± .011	"	31.56	35.17	"	"	"	"	.950 ± 0.011
22	24	1.200 ± .011	"	38.04	42.14	"	"	"	"	1.150 ± 0.011
23	25	1.250 ± .011	"	39.67	44.24	"	"	"	"	1.200 ± 0.011
24	26	1.300 ± .011	"	41.26	46.01	"	"	"	"	1.250 ± 0.011
25	34	1.700 ± .015	"	54.18	60.45	"	"	"	"	1.650 ± 0.015
26	37	1.850 ± .015	"	58.71	65.48	"	"	"	"	1.800 ± 0.015
27	40	2.000 ± .015	"	63.88	71.29	"	"	"	"	1.950 ± 0.015
28	50	2.500 ± .015	"	80.04	89.37	"	"	"	"	2.450 ± 0.015
29	60	3.000 ± .020	"	96.20	107.41	"	"	"	"	2.950 ± 0.015
30	64	3.200 ± .020	"	101.56	113.26	"	"	"	"	3.150 ± 0.015
31	9	.450 ± .010	.038 ± .003	19.21	20.64	24	.024	7/32	25,4	.400 ± 0.007
32	10	.500 ± .010	"	20.84	22.54	"	"	"	"	.450 ± 0.007
33	14	.700 ± .010	"	29.48	31.95	"	"	"	"	.650 ± 0.007
34	15	.750 ± .010	"	31.96	34.70	"	"	"	"	.700 ± 0.007
35	16	.800 ± .011	"	33.80	36.65	"	"	"	"	.750 ± 0.011
36	20	1.000 ± .011	"	42.44	46.05	"	"	"	"	.950 ± 0.011
37	24	1.200 ± .011	"	51.64	55.94	"	"	"	"	1.150 ± 0.011
38	25	1.250 ± .011	"	53.27	57.84	"	"	"	"	1.200 ± 0.011
39	26	1.300 ± .011	"	55.40	60.15	"	"	"	"	1.250 ± 0.011
40	34	1.700 ± .015	"	72.68	78.95	"	"	"	"	1.650 ± 0.015
41	37	1.850 ± .015	"	78.84	85.60	"	"	"	"	1.800 ± 0.015
42	40	2.000 ± .015	"	85.64	93.05	"	"	"	"	1.950 ± 0.015
43	50	2.500 ± .015	"	107.24	116.57	"	"	"	"	2.450 ± 0.015
44	60	3.000 ± .020	"	128.84	140.05	"	"	"	"	2.950 ± 0.015
45	64	3.200 ± .020	"	136.37	148.06	"	"	"	"	3.150 ± 0.015

1/ Nominal values are for information only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Insulation material: In accordance with table I of MIL-C-49055.

Voltage rating: 300 volts rms at sea level.

Conductor coating: Bare.

Conductor stranding: See table I.

Weight: See table I.

Insulation flaws: 2,500 volts rms.

Dielectric withstanding voltage: 2,000 volts rms.

Conductor resistance: See table I.

Thermal shock: -55°C ±5°C to 150°C ±5°C.

Flammability: Nonburning.

Ⓐ Flexing endurance: -55°C ±5°C to 150°C ±5°C for 24 AWG, 100 cycles; 26 AWG, 300 cycles; and 28 AWG, 500 cycles.

Moisture resistance:
Insulation resistance 50 megohms: 1,000 feet (minimum).

Low pressure (elevated temperature): 150°C ±5°C (maximum conductor temperature).

Ⓐ Markings: Not applicable.

Ⓐ Strippability: Not applicable.

Ⓐ Solderability: Not applicable.

Ⓐ Tear groove propagation: Not applicable.

Part number: M49055/11 (dash number from table I herein followed by material designator from table I of MIL-C-49055).

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AV, MI, AR
Navy - EC, OS, SH
Air Force - 11, 17, 99
DLA - IS, ES
NSA
NAS
NASA - MSFC

User activities:
Army - ME, AT
Navy - MC

Agent:
DLA - ES

Preparing activity:
Army - CR

(Project 6145-0859-1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Communications- Electronics Command
and Fort Monmouth
ATTN: DRSEL-ED-TO
Fort Monmouth, New Jersey 07703

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49055/11A	Cables, Special Purpose, Electrical, Flexible, Flat, Unshielded, Round Conductor

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:46Z
Check the source to verify that this is the current version before use.